

Module: ITNPBD2 Spring 2026

Assessment: Main assignment

Due Date/Time: 13/03/2026

AIAS Levels Allowed: 2

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ITNPBD2 – Data Analysis Report

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Module : ITNPBD2 – Representing and Manipulating Data

Semester : Spring 2026

Dataset : JCPenney Product and Review Dataset

1. Introduction

Data analysis is a critical aspect in understanding customer behaviour and product performance on e-commerce sites. For any retail business, data is generated on various parameters such as product data, customer review data, and user data. Analysis of such data helps organisations understand how well a product is performing, how customers rate it, and how customer behaviour varies from user to user.

The aim and objective of this assignment are to analyse the provided JCPenney data set and extract useful information such as customer ratings, product performances, and how customer reviews are related to product prices using Python and its pandas library.

The appendix contains the entire Jupyter Notebook used to analyse the data.

2. Dataset Description

The dataset consists of three main data sources:

2.1 Products Dataset

This dataset contains information about products sold on the platform.

Attributes include:

Uniq_id Unique identifier for each product

SKU Stock keeping unit identifier

Name Product name

Description Product description

Price Price of the product

Avg_Score Average rating score

2.2 Reviews Dataset

This dataset contains customer reviews for products.

Uniq_id Product identifier

Username The customer who wrote the review

Score Rating given by the customer

Review Review text

2.3 Users Dataset

This dataset contains information about the users who wrote reviews.

Username Unique username

DOB Date of birth

State Location of the user

3. Data Preparation and Cleaning

Before the analysis, the datasets were cleaned and preprocessed. Cleaning of the data is an important aspect of any data analysis problem since real-world datasets may have missing or inconsistent values.

The following cleaning operations were performed:

Handling of Missing Values

The columns “SKU” and “Description” had missing values. These missing values were handled using the following methods:

- Missing values in the Price column were replaced with the mean value.
- Missing values in the SKU column were replaced with “Unknown”.
- Missing values in the Description column were replaced with “No Description”.

The datasets were now free of missing values.

Data Type Conversion

The following columns were converted to the respective data types:

- The Price column was converted to a numeric data type.
- The date of birth column was converted to a datetime data type.

Merging of Datasets

To carry out the analysis, the three datasets were merged based on common attributes.

1. The Users and Reviews datasets were merged using the “Username” column.
2. The merged datasets were then merged with the Products dataset using the “Uniq_id” column.

The resulting datasets contained all the necessary information.

4. Data Analysis

Various analyses were conducted to gain insights from the data set.

4.1 Product Price Distribution

The distribution of the product prices was analysed by using a histogram.

It is observed that most products are in a lower price range. However, a few products are significantly higher-priced. This kind of distribution is right-skewed.

This kind of distribution is common in a retail data set because there are products that are more expensive and products that are less expensive.

The distribution of product prices helps businesses evaluate the pricing strategy.

4.2 Distribution of Ratings

The customer ratings were analysed to gain insights into overall customer satisfaction.

A bar chart is created to show the ratings.

The results show that the ratings are higher (4 and 5 stars) compared to the lower ratings.

This implies that the customers are satisfied with the products available in the data set.

4.3 Product Rating Analysis

Another kind of analysis is conducted to see how many products are rated.

The difference between the previous analysis and the present one is the number of products reviewed.

For instance:

- Even if a product has a high number of reviews with a rating of 5, it will only be counted as one product during analysis.
- Therefore, it will be possible to measure the performance of the product rather than the number of reviews.

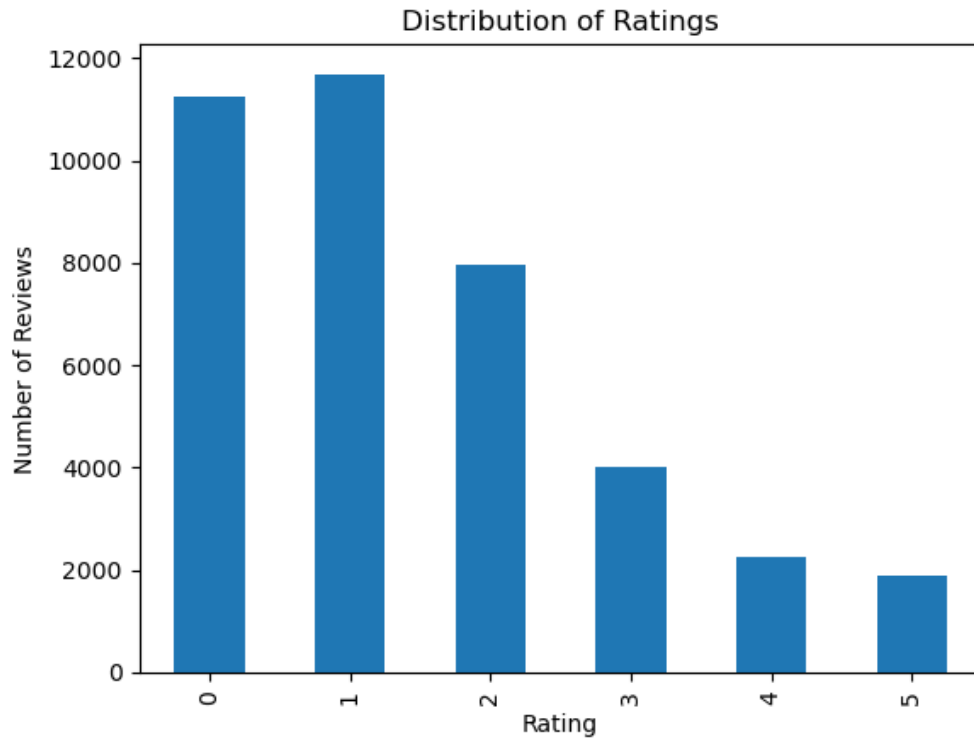
4.4 Top-Rated Products

- The analysis of the given dataset also helped in finding products with the highest average rating.
- The grouping of the data based on the product name and finding the mean helped in finding the products with the highest average rating in the given data.
- These products show high levels of satisfaction among customers and may represent high-quality products in the store.

5. Insights and Findings

From the analysis of the given problem, it can be stated that:

1. The price range of the products in the given data is heavily skewed.
2. The overall rating of the customers in the given data is positive.
3. High numbers of products have high ratings in the given data. Therefore, it can be stated that the overall quality of the products in the given data is positive.
4. The combined analysis of user, product, and review will help in a deeper analysis of the customers.



The distribution chart of the rating indicates that the majority of the product reviews in the database are concentrated in the lower rating categories, i.e., rating 0 and rating 1. As the value of the rating increases, the number of product reviews available in the database decreases. Ratings 4 and 5 have a lower occurrence in the database. It appears that a majority of the products in the database have lower ratings compared to those with higher ratings. This may imply that the satisfaction of customers with products varies. It may also imply the need to find products with lower ratings and work on improving the quality of the products.

Conclusion

The aim of this project is to show the power of Python and data analysis in extracting useful insights from available retail-related data. By using the available product-related data, we are able to extract useful insights.

The results show that most products have good ratings, with an irregular distribution of prices among the products. These results can be used to make intelligent decisions regarding product management and how to increase customer satisfaction.

The project can be taken further by adding new variables to the analysis.

Representing and Manipulating Data Assignment

Importing libraries which are required

1. Business Understanding

What we're aiming for: We want to get a good understanding of how JC Penney customers feel about the products and how those products actually perform based on the review data.

What we're looking at: By reviewing the ratings, price points, and frequency of reviews, we want to find trends that can help steer the business in a positive direction in terms of customer satisfaction.

What we're looking for:

- What types of products have the highest and lowest ratings?
- What types of products have the most reviews?
- Do products with a higher price point have a higher rating?
- What does the rating distribution say about overall customer satisfaction?
- What products need work based on the reviews?

2. Data Understanding

The data is scattered across several files, and everything is linked by who has written what and what the products are.

Files used in this analysis:

- products.csv, where the fundamental data about the products is stored, including the name, category, price, and a small description.
- reviews.csv, where the opinions from the customers are stored, along with the scores they provided for each review.
- users.csv, where information about the users is available.
- jcpenny_products.json, where additional information is available for the products.
- jcpenny_reviewers.json, where information about the users who reviewed the products is available in JSON format.

Explanation of how these pieces fit together:

- Uniq_id is the link between the products and the reviews for each product.
- Username is the link between users and the reviews written by them.

What we did initially:

- We examined the data structures.

- We examined if there were any missing values.
- We examined the data types for each column.
- We generated descriptive statistics.

Importing Libraries

```
import pandas as pd
import numpy as np
import json
import matplotlib.pyplot as plt

import os
os.getcwd() #to get our current working directory we use getcwd() function

'c:\\Users\\aksha\\OneDrive\\Desktop\\D2_Assignment'
```

importing CSV files

```
products = pd.read_csv(r"JCPenneyFiles\JCPenneyFiles\products.csv")
#to get/read products.csv file into out code
reviews = pd.read_csv(r"JCPenneyFiles\JCPenneyFiles\reviews.csv") #to get/read reviews.csv file into out code
users = pd.read_csv(r"JCPenneyFiles\JCPenneyFiles\users.csv") #to get/read users.csv file into out code
```

Getting information about CSV Files

```
print("Products dataset shape:", products.shape)
Products dataset shape: (7982, 6)

print("Reviews dataset shape:", reviews.shape)
Reviews dataset shape: (39063, 4)

print("Users dataset shape:", users.shape)
Users dataset shape: (5000, 3)

products_df = pd.read_json(
    r"JCPenneyFiles\JCPenneyFiles\jcpenny_products.json",
    lines=True
)
reviewers_df = pd.read_json(
    r"JCPenneyFiles\JCPenneyFiles\jcpenny_reviewers.json",
    lines=True
)
products_df.head() #to get first 5 rows of products csv file
```

	uniq_id	sku	\
0	b6c0b6bea69c722939585baeac73c13d	pp5006380337	
1	93e5272c51d8cce02597e3ce67b7ad0a	pp5006380337	
2	013e320f2f2ec0cf5b3ff5418d688528	pp5006380337	
3	505e6633d81f2cb7400c0cfa0394c427	pp5006380337	
4	d969a8542122e1331e304b09f81a83f6	pp5006380337	

	name_title	\
0	Alfred Dunner® Essential Pull On Capri Pant	
1	Alfred Dunner® Essential Pull On Capri Pant	
2	Alfred Dunner® Essential Pull On Capri Pant	
3	Alfred Dunner® Essential Pull On Capri Pant	
4	Alfred Dunner® Essential Pull On Capri Pant	

	description	list_price
0	You'll return to our Alfred Dunner pull-on cap...	41.09
24.16		
1	You'll return to our Alfred Dunner pull-on cap...	41.09
24.16		
2	You'll return to our Alfred Dunner pull-on cap...	41.09
24.16		
3	You'll return to our Alfred Dunner pull-on cap...	41.09
24.16		
4	You'll return to our Alfred Dunner pull-on cap...	41.09
24.16		

	category	category_tree	average_product_rating
0	alfred dunner	jcpenny women alfred dunner	2.625
1	alfred dunner	jcpenny women alfred dunner	3.000
2	view all	jcpenny women view all	2.625
3	view all	jcpenny women view all	3.500
4	view all	jcpenny women view all	3.125

	product_url	\
0	http://www.jcpenny.com/alfred-dunner-essentia...	
1	http://www.jcpenny.com/alfred-dunner-essentia...	
2	http://www.jcpenny.com/alfred-dunner-essentia...	
3	http://www.jcpenny.com/alfred-dunner-essentia...	
4	http://www.jcpenny.com/alfred-dunner-essentia...	

	product_image_urls	brand	\
0	http://s7d9.scene7.com/is/image/JCPenny/DP122...	Alfred Dunner	
1	http://s7d9.scene7.com/is/image/JCPenny/DP122...	Alfred Dunner	

```

2 http://s7d9.scene7.com/is/image/JCPenney/DP122... Alfred Dunner
3 http://s7d9.scene7.com/is/image/JCPenney/DP122... Alfred Dunner
4 http://s7d9.scene7.com/is/image/JCPenney/DP122... Alfred Dunner

```

```

total_number_reviews
Reviews \
0      8  [{'User': 'fsdv4141', 'Review': 'You never
hav...
1      8  [{'User': 'tpcu2211', 'Review': 'You never
hav...
2      8  [{'User': 'pcfg3234', 'Review': 'You never
hav...
3      8  [{'User': 'ngrq4411', 'Review': 'You never
hav...
4      8  [{'User': 'nbmi2334', 'Review': 'You never
hav...

```

```

Bought With
0 [898e42fe937a33e8ce5e900ca7a4d924, 8c02c262567...
1 [bc9ab3406dcaa84a123b9da862e6367d, 18eb69e8fc2...
2 [3ce70f519a9cfdd85cdbdec358e5347, b0295c96d2b...
3 [efcd811edccbeb5e67eaa8ef0d991f7c, 7b2cc00171e...
4 [0ca5ad2a218f59eb83eec1e248a0782d, 9869fc8da14...

```

```
reviewers_df.head() #to get first 5 rows of reviewers
```

```

Username      DOB      State \
0 bkpn1412  31.07.1983      Oregon
1 gqjs4414  27.07.1998  Massachusetts
2 eehe1434  08.08.1950      Idaho
3 hkxj1334  03.08.1969      Florida
4 jjbd1412  26.07.2001      Georgia

```

```

Reviewed
0 [cea76118f6a9110a893de2b7654319c0]
1 [fa04fe6c0dd5189f54fe600838da43d3]
2 []
3 [f129b1803f447c2b1ce43508fb822810, 3b0c9bc0be6...
4 []

```

```
users.head() #to get first 5 rows of users file
```

```

Username      DOB      State
0 bkpn1412  31.07.1983      Oregon
1 gqjs4414  27.07.1998  Massachusetts
2 eehe1434  08.08.1950      Idaho
3 hkxj1334  03.08.1969      Florida
4 jjbd1412  26.07.2001      Georgia

```

```
products.info() #to get structure of products file
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7982 entries, 0 to 7981
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Uniq_id         7982 non-null   object
1   SKU             7915 non-null   object
2   Name            7982 non-null   object
3   Description      7439 non-null   object
4   Price           5816 non-null   float64
5   Av_Score        7982 non-null   float64
dtypes: float64(2), object(4)
memory usage: 374.3+ KB
```

`reviews.info()` *#to get structure of reviews csv file*

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 39063 entries, 0 to 39062
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Uniq_id         39063 non-null   object
1   Username        39063 non-null   object
2   Score           39063 non-null   int64
3   Review          39063 non-null   object
dtypes: int64(1), object(3)
memory usage: 1.2+ MB
```

`users.info()` *#to get structure of users csv file*

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Username        5000 non-null   object
1   DOB             5000 non-null   object
2   State           5000 non-null   object
dtypes: object(3)
memory usage: 117.3+ KB
```

`products.describe()` *#to get description of products file*

	Price	Av_Score
count	5816.000000	7982.000000
mean	144.776618	2.988683
std	499.223719	0.911673
min	-65.270000	1.000000
25%	40.700000	2.500000
50%	58.010000	3.000000

75%	87.020000	3.500000
max	17122.170000	5.000000

```
reviews.describe() #to get description of reviews file
```

	Score
count	39063.000000
mean	1.487648
std	1.400332
min	0.000000
25%	0.000000
50%	1.000000
75%	2.000000
max	5.000000

```
users.describe() #to get description of users file
```

	Username	DOB	State
count	5000	5000	5000
unique	4999	52	57
top	dqft3311	07.08.1953	Massachusetts
freq	2	112	107

Checking Missing Values

```
products.isnull().sum() #checking how many are missing values in products csv file
```

Uniq_id	0
SKU	67
Name	0
Description	543
Price	2166
Av_Score	0
dtype:	int64

3. Data Preparation

Filling the Missing values

The missing information in the data was handled in a manner that ensures proper analysis. This was done by setting the information in the SKU column to "Unknown," the information in the Description column to "No Description," and the missing information in the Price column to the average price.

```
# products = products.dropna()
# or we can also fill the missing values with mean imputation like
products["SKU"] = products["SKU"].fillna("Unknown") #filling missing values in SKU column with "Unknown"
```

```

products["Description"] = products["Description"].fillna("No
Description") #filling missing values in Description column with "No
Description"
products["Price"] = products["Price"].fillna(products["Price"].mean())
#filling missing values in Price column with mean

products.isnull().sum() #veryfing that there are no missing values in
products csv file

Uniq_id      0
SKU          0
Name         0
Description   0
Price        0
Av_Score     0
dtype: int64

reviews.isnull().sum() #checking how many are missing values in
reviews csv file

Uniq_id      0
Username     0
Score        0
Review       0
dtype: int64

users.isnull().sum() #checking how many are missing values in users
csv file

Username     0
DOB          0
State        0
dtype: int64

products["Price"].describe() #this is the description after filling
the missing values from products csv file

count      7982.000000
mean       144.776618
std        426.129318
min        -65.270000
25%         47.300000
50%         75.640000
75%        144.776618
max        17122.170000
Name: Price, dtype: float64

```

Merging Tables

The tables are joined to combine related data from different sources. One table has product data, while the other has customer ratings and feedback data. By joining these two tables using

a common key, we are able to link product data with customer ratings data under the common column of Uniq_id.

```
merged = pd.merge(products, reviews, on="Uniq_id")
merged.groupby("Name")["Score"].mean() #to get the average rating for each product
```

```
Name
1 CT. Certified Diamond Solitaire Ring
2.333333
1 CT. T.W. Certified Diamond 14K White Gold Bridal Ring Set
5.000000
1 CT. T.W. Certified Diamond 14K White Gold Princess-Cut Band
1.000000
1 CT. T.W. Certified Diamond 14K Yellow Gold Bridal Ring Set
1.000000
1 CT. T.W. Diamond 10K White Gold Cluster Ring
0.000000
...

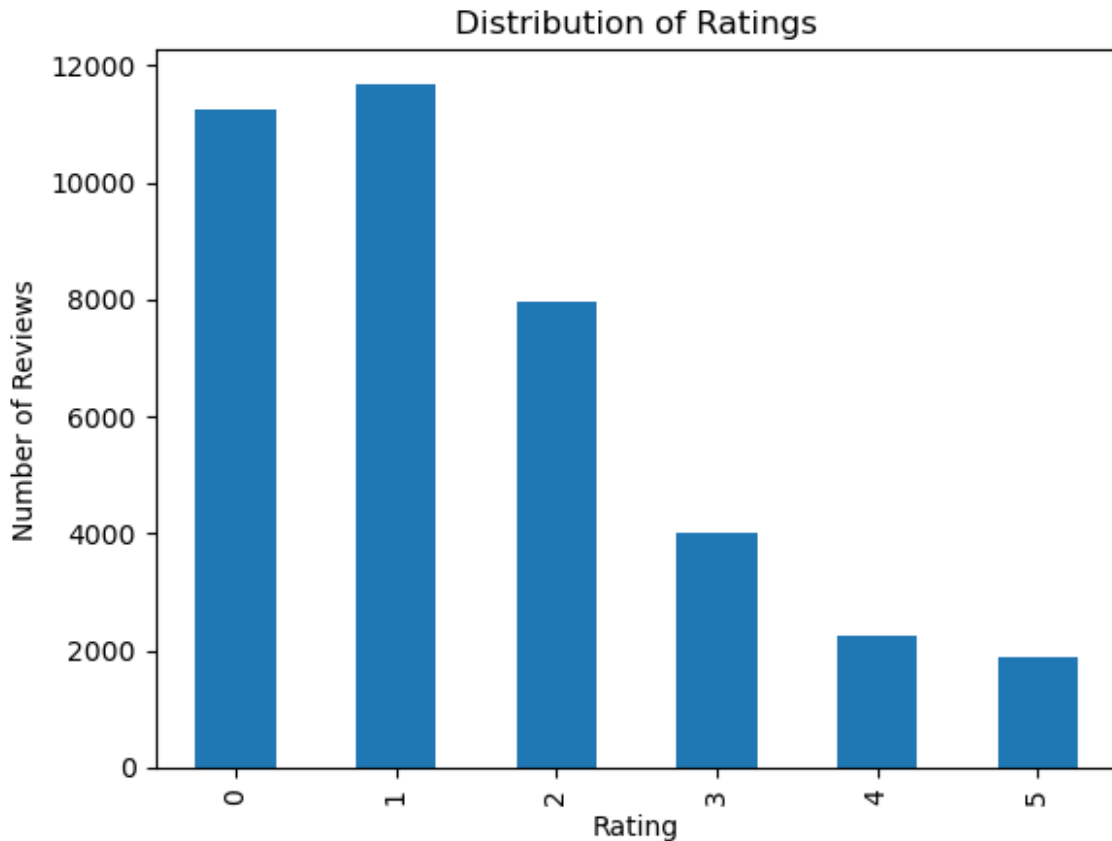
1/4 CT. T.W. White & Color-Enhanced Black Diamond Weave Ring
2.000000
1/2 CT. Princess Certified Diamond Solitaire Ring
1.000000
1/2 CT. T.W. Diamond 10K Yellow Gold Contoured Anniversary Band
0.750000
1/2 CT. T.W. Diamond Bridal Set
0.000000
1/3 CT. T.W. Diamond 3-Stone Promise Ring
1.000000
Name: Score, Length: 6001, dtype: float64
```

```
users_reviews = pd.merge(users, reviews, on="Username")
users.head() #to get first 5 rows of users csv file
```

	Username	DOB	State
0	bkpn1412	31.07.1983	Oregon
1	gqjs4414	27.07.1998	Massachusetts
2	eehe1434	08.08.1950	Idaho
3	hkxj1334	03.08.1969	Florida
4	jjbd1412	26.07.2001	Georgia

4. Insight Analysis

```
merged["Score"].value_counts().sort_index().plot(kind="bar")
plt.xlabel("Rating")
plt.ylabel("Number of Reviews")
plt.title("Distribution of Ratings")
plt.show()
```



```
rating_summary = merged.groupby("Score")
["Name"].nunique().reset_index() #it gives how many unique products
received each rating at least once
rating_summary.columns = ["Rating", "Number_of_Products"]
rating_summary
```

	Rating	Number_of_Products
0	0	4146
1	1	4243
2	2	3612
3	3	2388
4	4	1637
5	5	1437

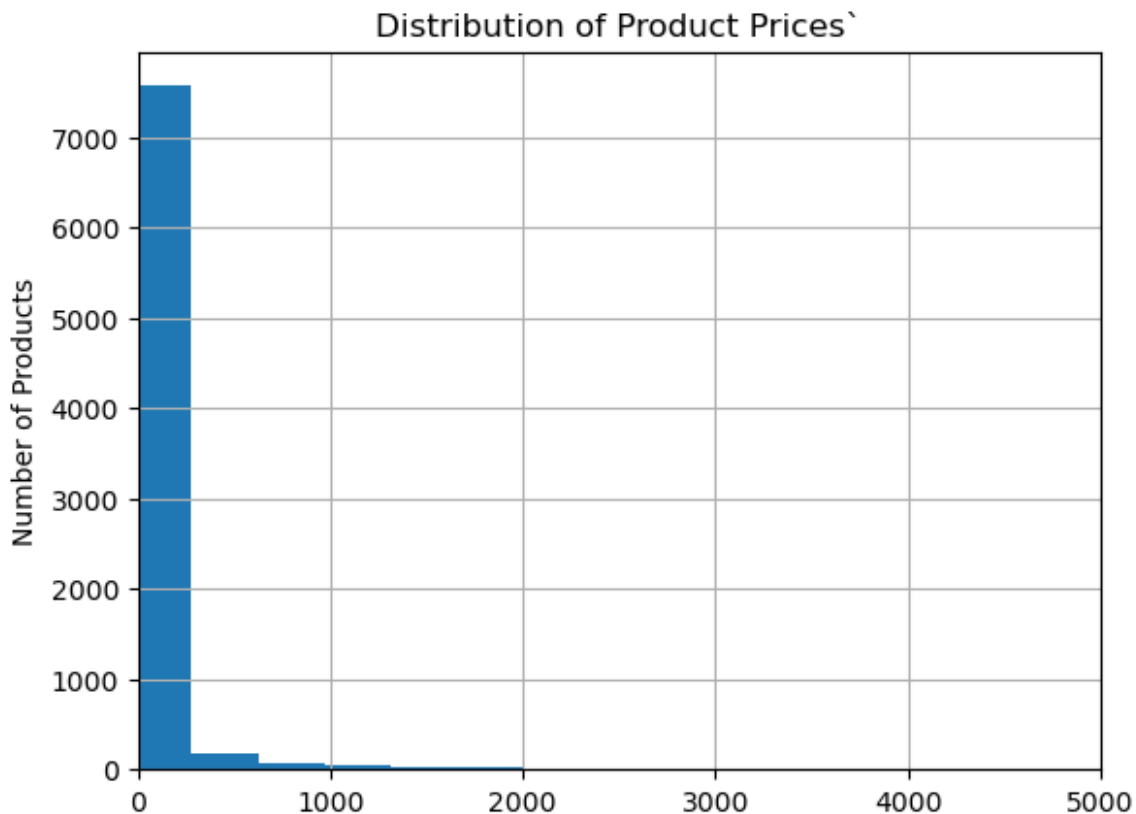
```
merged.sort_values(by="Price", ascending=False)[["Name",
"Price"]].head()
```

	Name	Price
28396	1½ CT. Round Certified Diamond Solitaire 14K Y...	17122.17
9317	Mens 1/2 CT. T.W. Diamond & Onyx 14K Gold Link...	13699.64
24634	1 CT. T.W. Diamond 14K White Gold Engagement Ring	9064.69
21335	Harmony Eternally in Love 3 CT. T.W. Certified...	8661.79
21329	Harmony Eternally in Love 3 CT. T.W. Certified...	8661.79


```
reviews["Uniq_id"].value_counts().head() #to get the top 5 products with most number of reviews
```

```
Uniq_id
7376425db74399716d23f1cc0c59684c    23
6c40fb8c09f9b94ef12003656c808cf1    23
2ce274384d293748bbf8bdebbc0aa145    23
867b08603fa032314919f0d7b5d47ad6    22
0db3122709238f7154dffalbaeedee3f    21
Name: count, dtype: int64
```

```
products["Price"].hist(bins=50)
plt.xlim(0, 5000)
plt.ylabel("Number of Products")
plt.title("Distribution of Product Prices`")
plt.show()
```



It shows how prices are distributed for the given set of products. The histogram shows how many items are within a certain range. The `plt.xlim(0, 5000)` is used to focus on a certain range of prices, from 0 to 5000. This range is chosen because prices are usually not extremely high. By selecting this prices range, we are essentially ignoring those high prices and only looking at the majority of prices. This is useful to understand how prices are distributed for the given set of products.

```
full_dataaa = pd.merge(users_reviews, products, on="Uniq_id") #merging
users and reviews data with products data to get a complete analysis
full_dataaa.head() #to get first 5 rows of the merged data
```

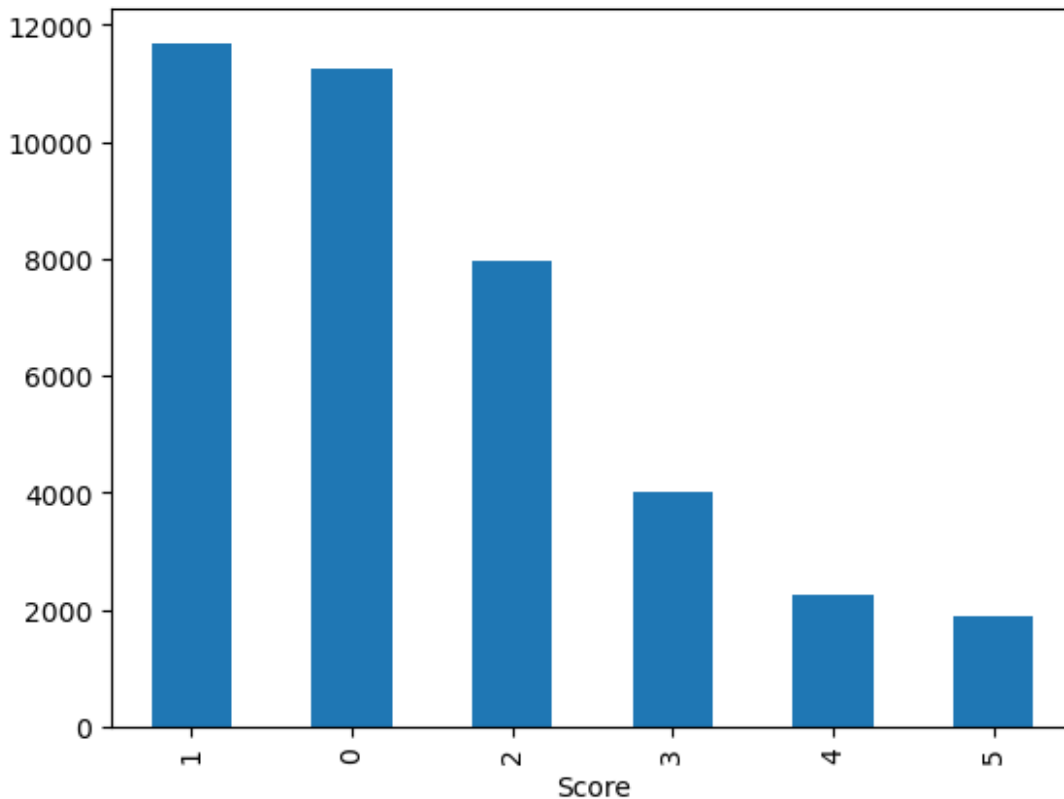
	Username	DOB	State	Uniq_id
Score \				
0	bkpn1412	31.07.1983	Oregon	2565bf02ec05a59cd1fd78955dda108d
0				
1	bkpn1412	31.07.1983	Oregon	e09b42d19cd350f9762f92d59ecf88ba
2				
2	bkpn1412	31.07.1983	Oregon	cea76118f6a9110a893de2b7654319c0
0				
3	bkpn1412	31.07.1983	Oregon	bc177e6724df06c676fd526e6d32461d
2				
4	bkpn1412	31.07.1983	Oregon	a0d0f5f772021d6320048dc7b61fb551
5				

	Review	SKU \
0	Love this heated mattress pad! Ordered the que...	pp5003160034
1	I am 511 and a size 6, with a very straight fi...	pp5006720496
2	I like the jacket accept it is thin not and th...	pp5005290257
3	Soft, comfortable, and it fits nice. Whats not...	pp5006950440
4	I bought the quick dri bath rug in the size 21...	pp5003380128

	Name \
0	Biddeford® Heated Mattress Pad
1	Worthington® Wide Waistband Cropped Pants - Tall
2	Silverwear™ Short-Sleeve Hooded Eyelet Jacket ...
3	Made for Life™ Mesh Print Tank Top or Shorts
4	JCPenney Home™ Ultra Soft Quick-Dri Bath Rug C...

	Description	Price
Av_Score		
0	Turn down the thermostat and turn up this comf...	144.776618
2.500000		
1	Enjoy the comfort of a modern fit and wide wai...	48.340000
2.428571		
2	Our short-sleeve French terry knit jacket is t...	52.030000
3.000000		
3	No Description	144.776618
3.375000		
4	Step into ultra soft luxury when you add these...	144.776618
3.000000		

```
reviews["Score"].value_counts().plot(kind="bar")
plt.show()
```



The bar chart illustrates how users in the data set have allocated their rating scores. From the bar chart, it is clear that a score of 1 is the most frequent, with a score of 0 following closely behind. This implies that these two scores appear more often in the data set compared to other scores. On the other hand, scores 2 and 3 appear to a lesser degree, while scores 4 and 5 appear least in the data set.

```
print("Median Product Price:", products["Price"].median())
```

```
Median Product Price: 75.64
```

```
# Creating the price the price labels to specify the catogory
price_ranges = pd.cut(products["Price"],
                        bins=[0,50,100,200,400,600,800,1000,5000],
                        labels=["1-50","51-100","101-200","201-400",
                              "401-600","601-800","801-1000","1000+"])
price_counts = price_ranges.value_counts().sort_index() # it gives the
number of products within each price range
```

```
print(price_counts)
```

```
Price
1-50      2223
51-100    2361
101-200   2791
201-400    245
```

```
401-600      79
601-800      55
801-1000     30
1000+       144
Name: count, dtype: int64
```

The median price was used to assess the price level in the data set. Unlike the average price, the median price does not fluctuate with high or low prices. Therefore, it is a more consistent measure of central tendency. Products within $\pm 10\%$ of the median were used to identify items that are within the normal range. These items are not likely to be impacted by high or low prices and are used to demonstrate how prices are set by the retailer.

```
topRated_products = merged.groupby("Name")
["Score"].mean().sort_values(ascending=False).head(10)
print(topRated_products)
```

```
Name
Fabric Possibilities Track-Arm 4-pc. Left-Arm Loveseat/Chaise
Sectional      5.0
Home Expressions™ Rosetti Euro Sham
5.0
T-Fal® Family Pro Deep Fryer
5.0
Collection by Michael Strahan Striped Black Suit Separates - Big &
Tall      5.0
Collection by Michael Strahan Reversible Textured-Buckle Belt
5.0
St. Johns Bay® Long-Sleeve Essential Crewneck T-Shirt- Petite
5.0
ZC0 Shape Bootcut Jeans - Plus
5.0
1 CT. T.W. Certified Diamond 14K White Gold Bridal Ring Set
5.0
Lugz® Empire Hi Mens Water-Resistant Hiking Boots
5.0
Classic Treasures™ Lab-Created Ruby and Diamond-Accent Drop Earrings
5.0
Name: Score, dtype: float64
```

Conclusion

This study delved into the JCPenney dataset using Python and various data processing tools such as Pandas and Matplotlib. After processing the data and handling the missing values and the union of the various datasets, the analysis could be conducted. The main idea of the analysis is that most of the items were within the mid-range price category. There were also only a few items that were priced extremely high.

With regard to the ratings of the items, the analysis revealed that the most common ratings were 1 and 0. On the other hand, ratings of 4 and 5 were less common. This could be an

indication of the quality of the products and the customer experience. The analysis also revealed which products were receiving the most reviews. This could be an indication of the exposure of the products. With the analysis of the data, various businesses could improve their sales by improving the quality of the products that were rated low and promoting the ones that were rated high.